МИНИСТЕРСТВО НАУКИ И ВЫСШЕГО ОБРАЗОВАНИЯ

РФ Федеральное государственное автономное

образовательное учреждение высшего образования

«МОСКОВСКИЙ ПОЛИТЕХНИЧЕСКИЙ УНИВЕРСИТЕТ»

Лабораторная работа 4 по дисциплине

«Автоматизация процессов жизненного цикла программных средств»

Выполнили:

Ст.гр.241-327

Кадерли Д.Б

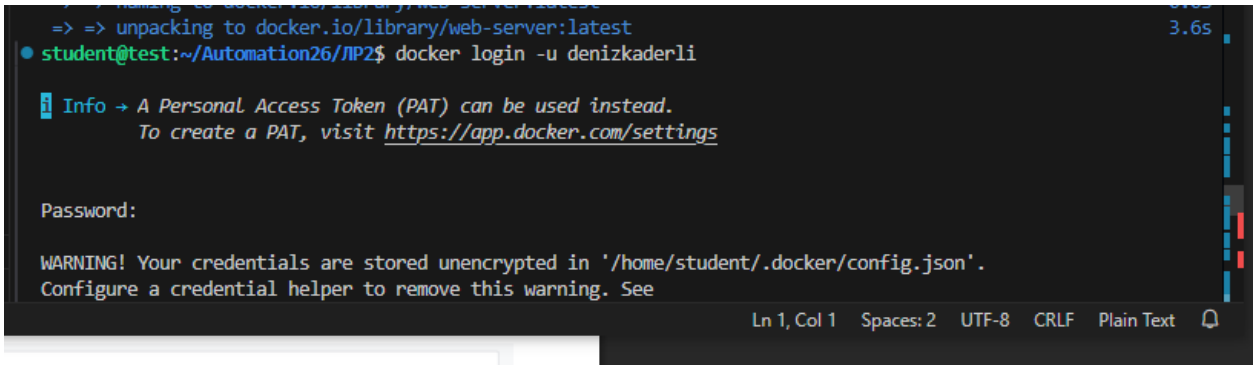
Гадукян Р.А

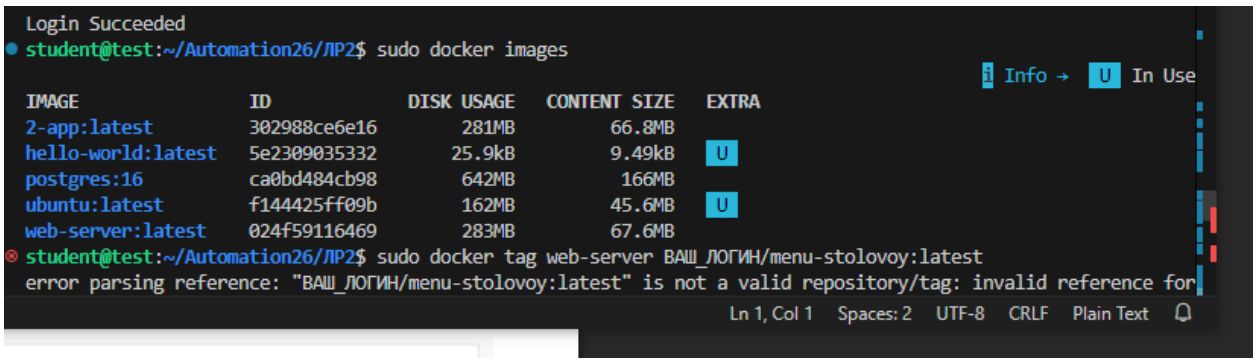
Алтайская М. С.

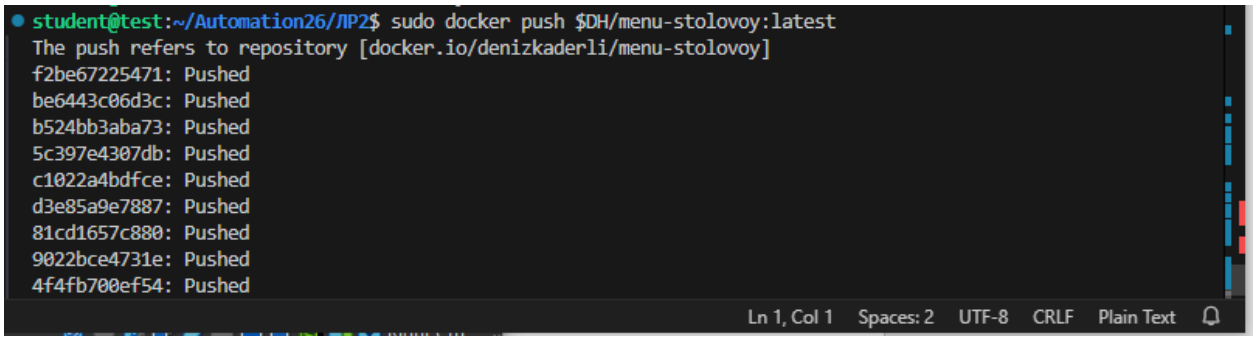
Москва - 2026

План выполнения лабораторной работы

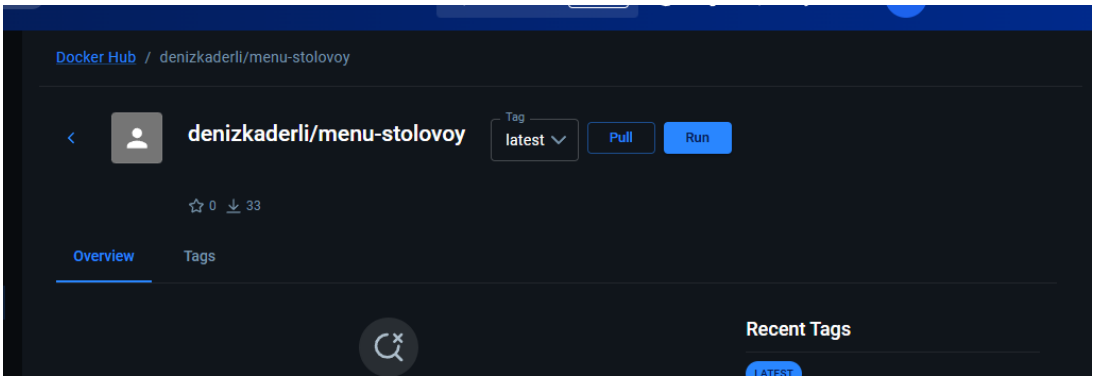
1. При помощи команды docker push собранный в ходе ЛР3 образ разрабатываемого приложения загрузить в Docker hub подключились к Dockerhub, вывели id наших контейнеров, выгрузили образ в репозиторий:



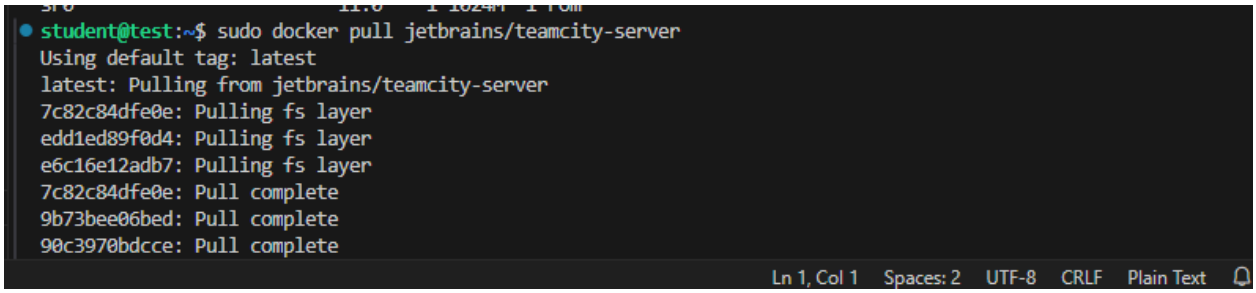


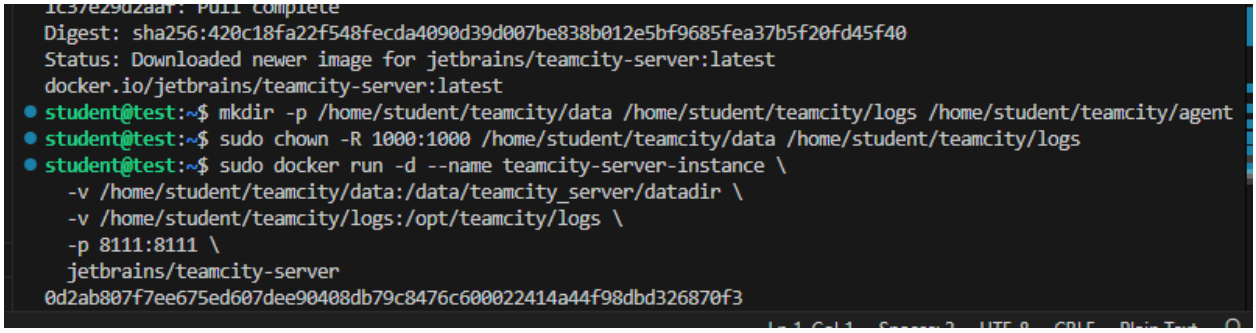


Образ успешно добавился:



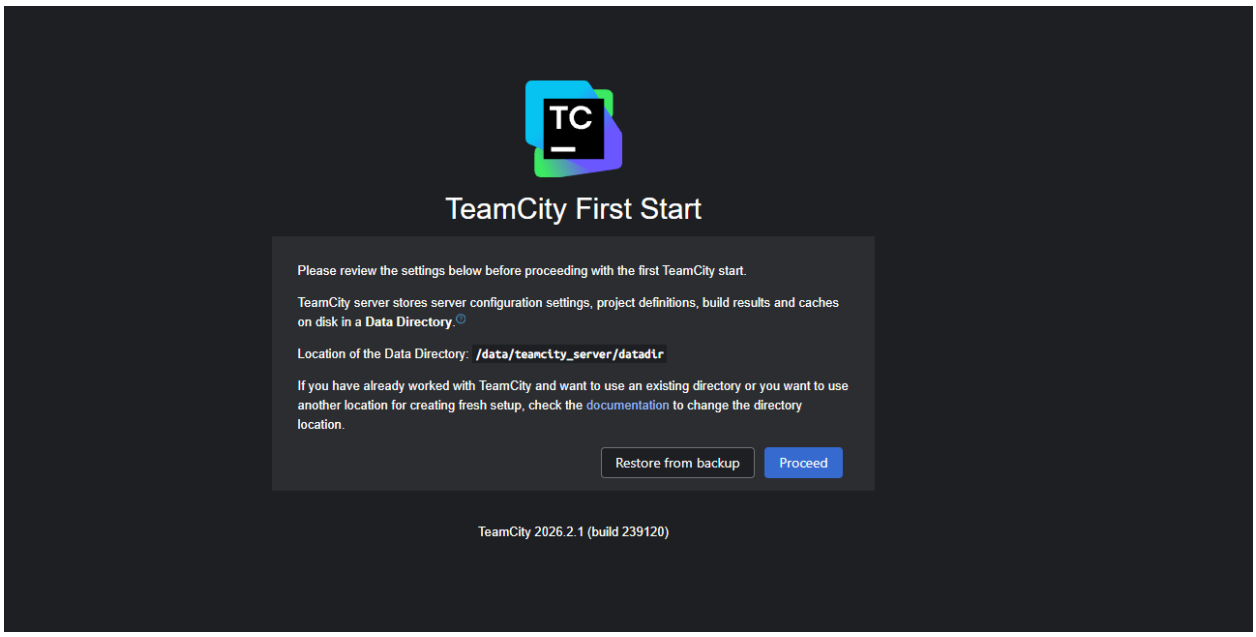
2. Загрузить с Docker Hub образ jetbrains/teamcity-server и развернуть его согласно инструкции на странице образа

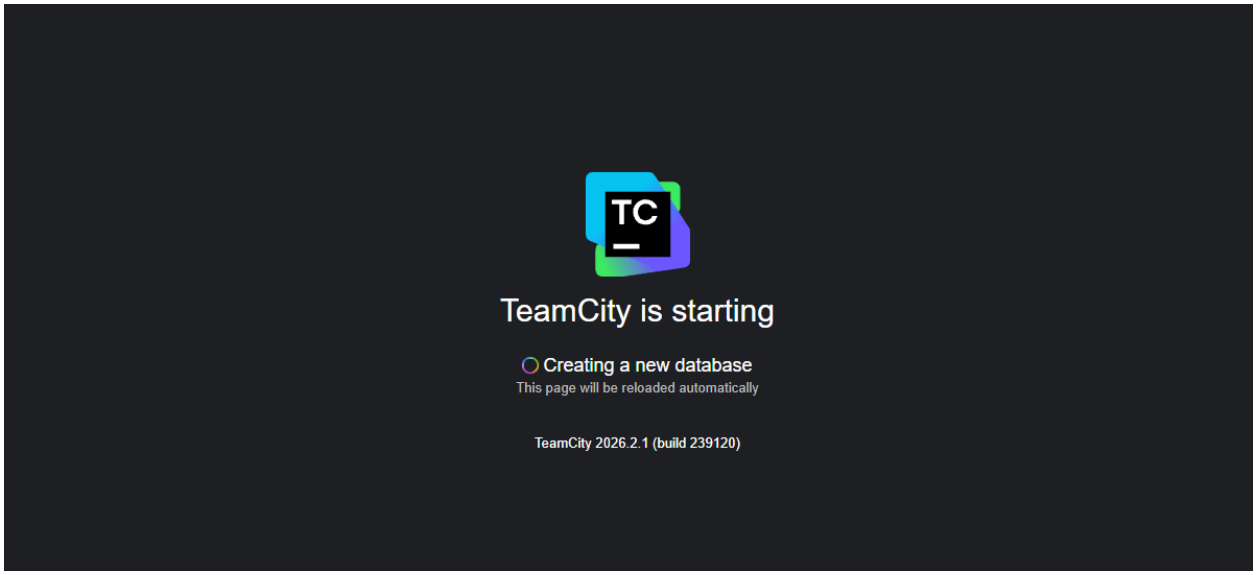






3. Открыть веб-интерфейс развернутого приложения, завести пользователя-администратора


TeamCity First Start
Please review the settings below before proceeding with the first TeamCity start.
TeamCity server stores server configuration settings, project definitions, build results and caches on disk in a Data Directory.
Location of the Data Directory: /data/teamcity_server/datadir
If you have already worked with TeamCity and want to use an existing directory or you want to use another location for creating fresh setup, check the documentation to change the directory location.
Restore from backup
Proceed
TeamCity 2026.2.1 (build 239120)


TeamCity is starting
Creating a new database
This page will be reloaded automatically
TeamCity 2026.2.1 (build 239120)



License Agreement

Version 1.3, effective as of November 27, 2025

Welcome to JetBrains TeamCity!

This TeamCity License Agreement constitutes a legally binding document, and it is important that You read it carefully.

You understand that by accepting this TeamCity (On-Premises) License Agreement (by clicking the "I agree" or a similar button or by accessing or using TeamCity), You enter into a legal agreement and agree to certain legal conditions for Yourself or for the legal entity that You represent.

By accepting this TeamCity License Agreement, You confirm that You understand it, You agree to it, and You are at least 13 years of age.

1. Introduction

This TeamCity License Agreement (**"Agreement"**) describes how You can access, purchase, and use TeamCity.

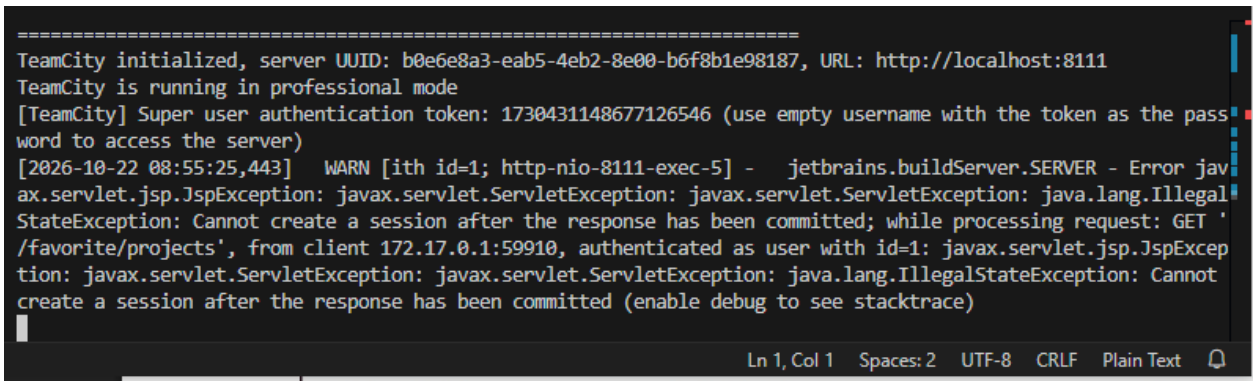
This Agreement is entered into between JetBrains s.r.o., a company with its registered office at Na Hřebenech II 1718/8, Nusle, 140 00 Prague 4, Czech Republic, ID No. 265 02 275, registered in the Commercial Register maintained by the Municipal Court in Prague, Section C, File No. 86211 (**"JetBrains"**, **"We"** , or **"Us"**) and You, either a legal entity or a natural person (**"You"**). JetBrains and You may each also be referred to individually as a **"Party"** or jointly as the **"Parties"**.

If You accept this Agreement on behalf of a legal entity, You confirm ('represent and warrant') that You are authorized to enter into agreements on behalf of that legal entity. If this Agreement is accepted using an email address provided by a legal entity, We will regard ('deem') You as authorized to represent that legal entity. You must be able to enter into contracts ('have capacity').

2. Definitions

a) Special legal phrases

There are certain phrases that have an accepted meaning for lawyers. To ensure this Agreement is clear and accessible, We have included the accepted 'legal' phrase in parentheses after the word to show that We intend it to have the accepted 'legal' meaning.

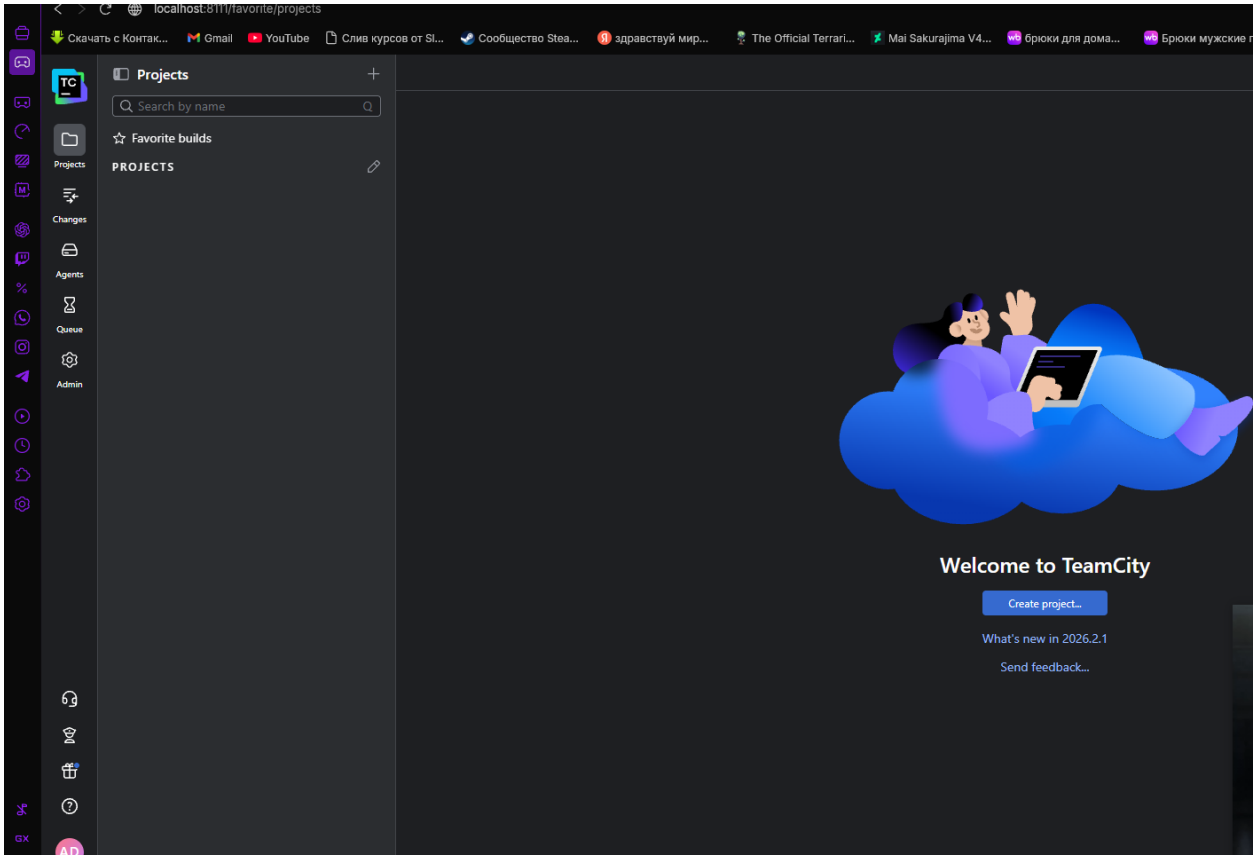


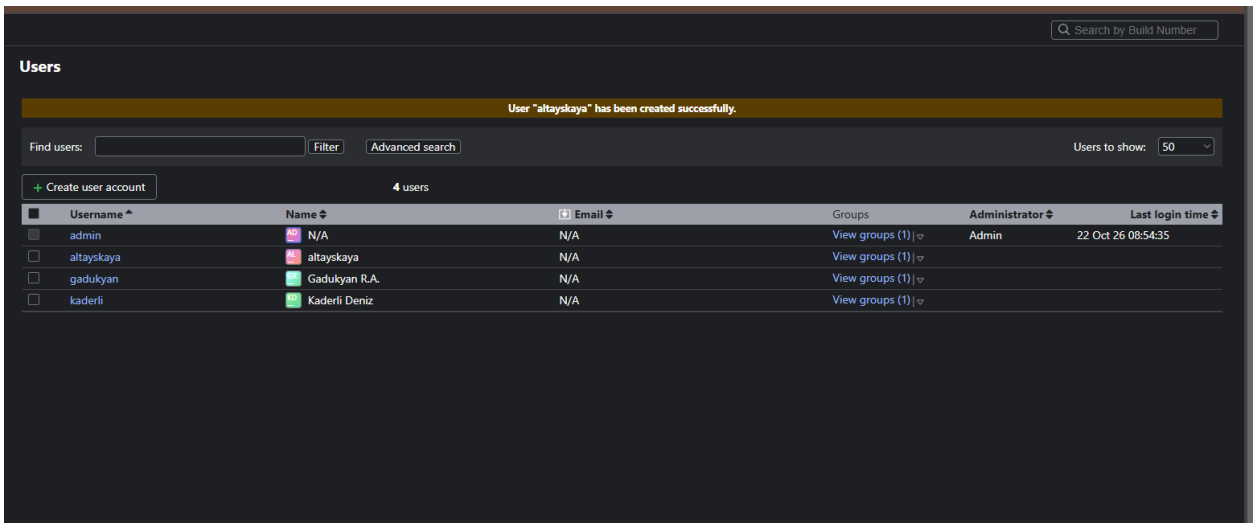
4. В панели администрирования завести по одному пользователю для каждого члена команды

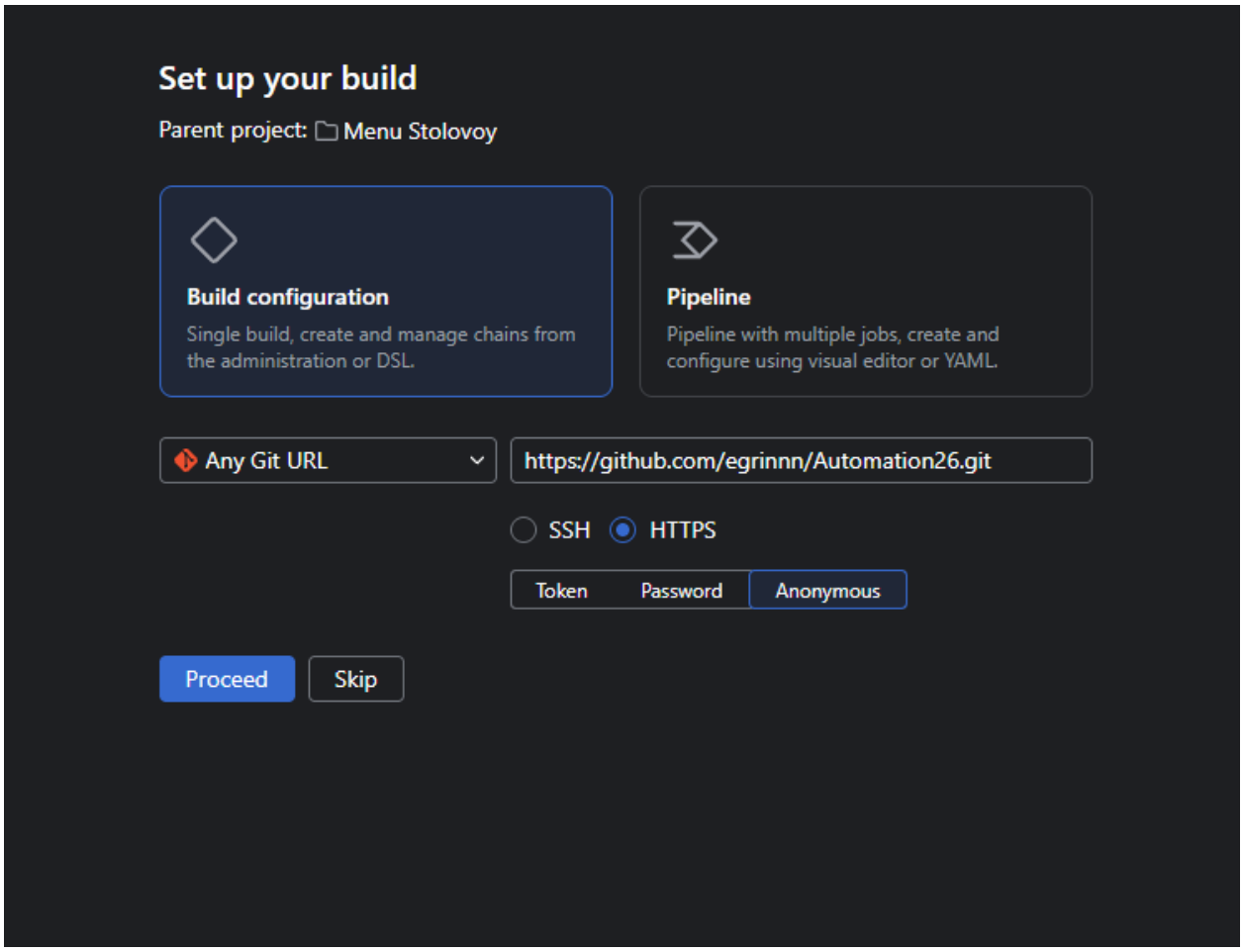
5. Завести репозиторий в TeamCity, привязав репозиторий проекта

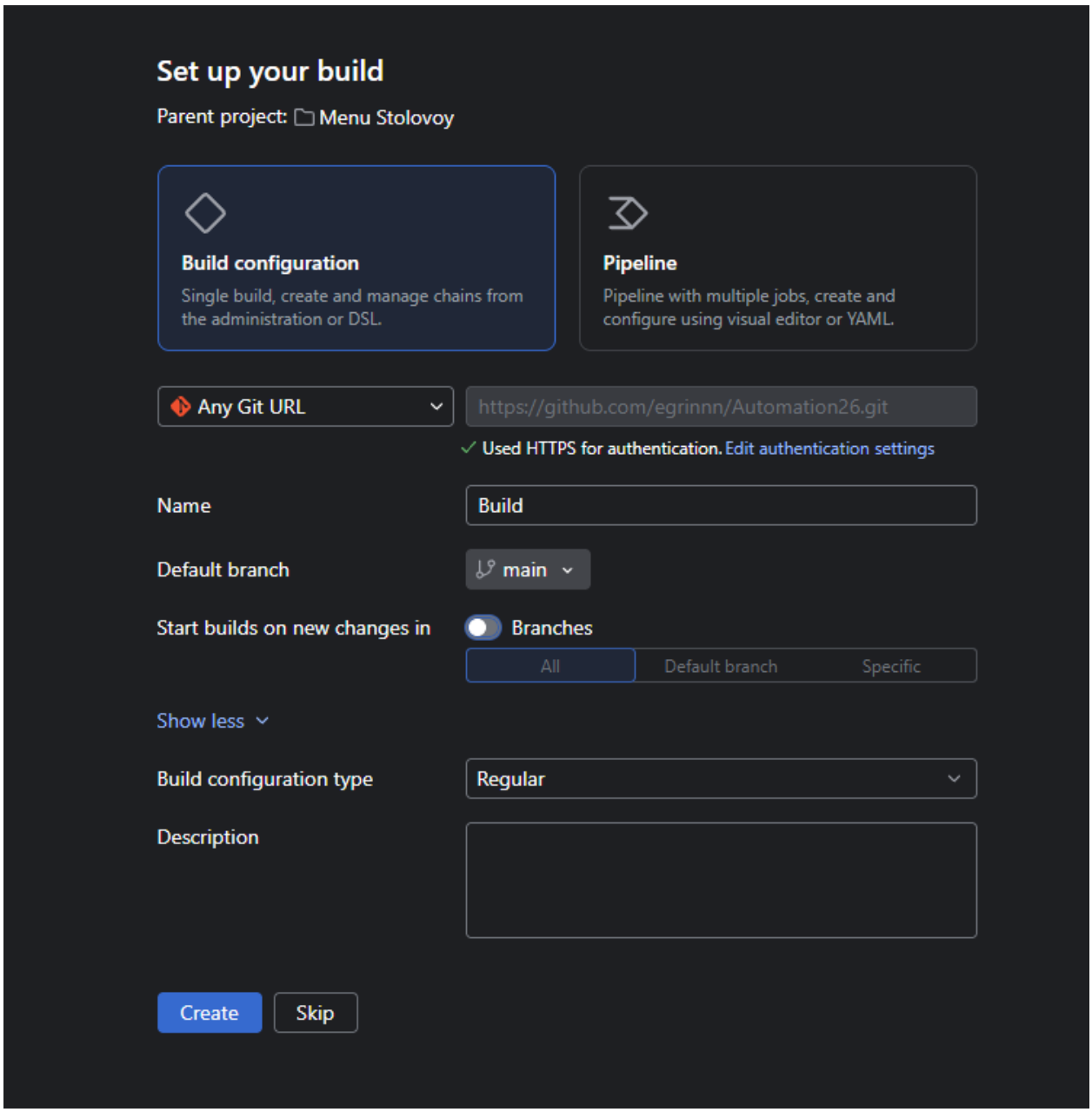
6. По выбору студента - либо из предложенных вариантов сборки, либо из реализованных студентом создать конфигурацию (последовательность шагов) для сборки разрабатываемого приложения в Docker образ и отправки его в Docker Hub

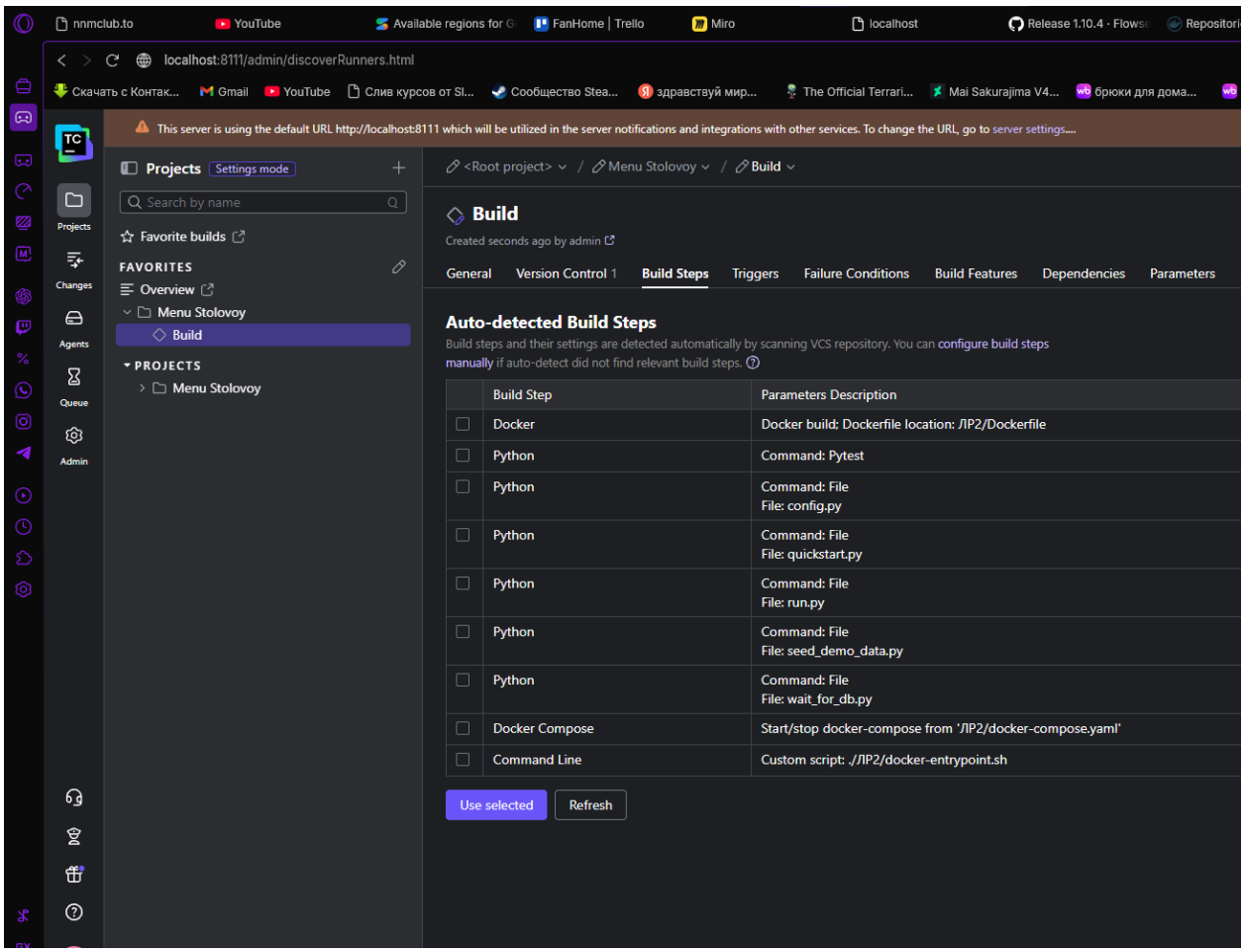
пункты 4 5 6 выполнены подряд и находятся ниже

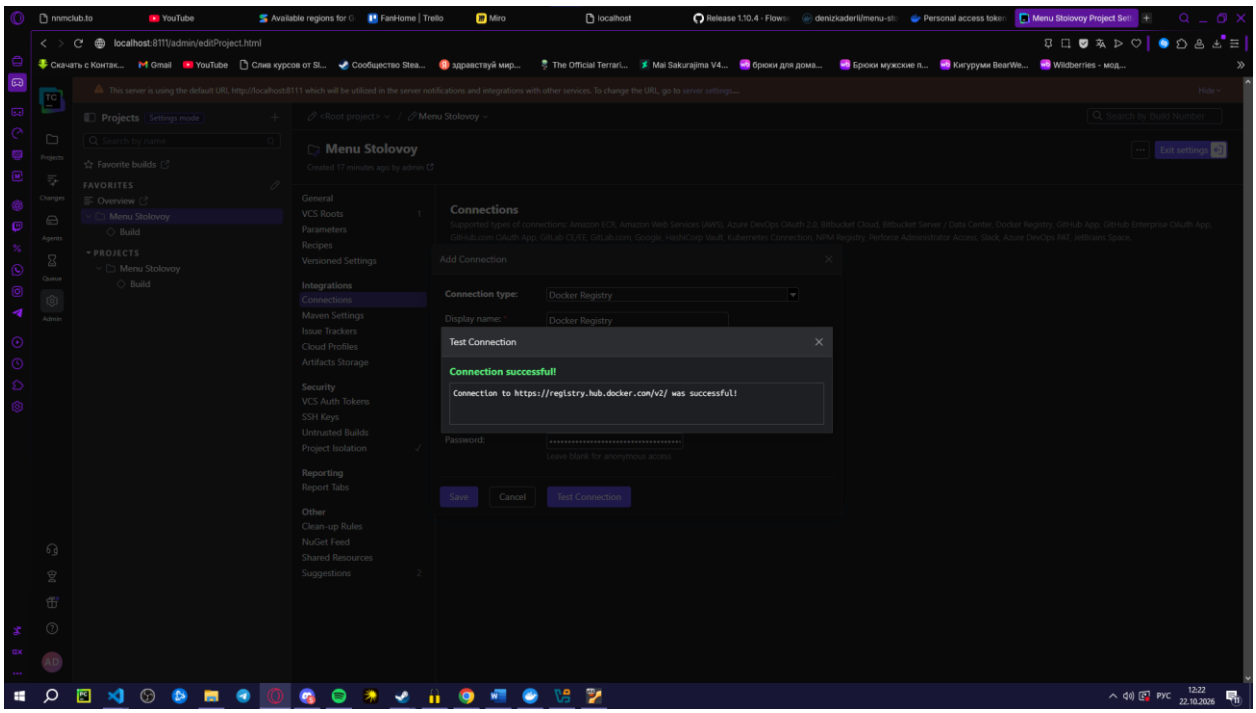


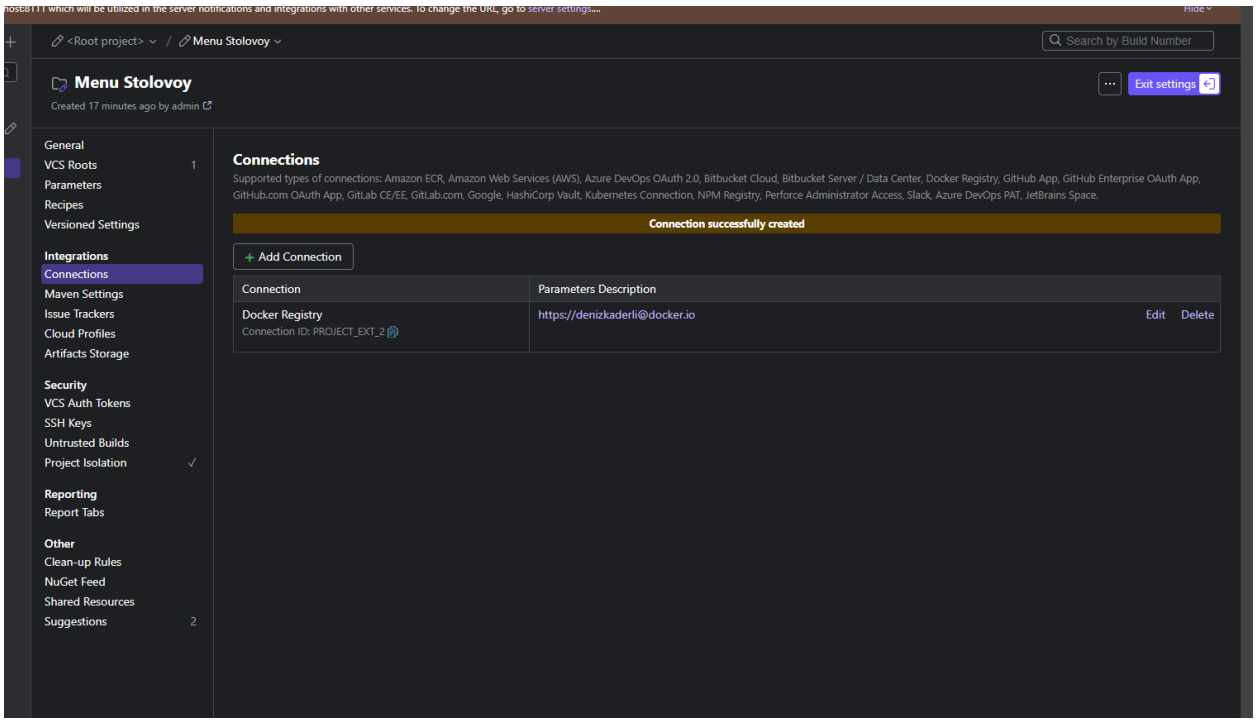

Search by Build Number
Users
User "altayskaya" has been created successfully.
Find users:
Filter
Advanced search
Users to show: 50
+ Create user account
4 users
Username
Name
Email
Groups
Administrator
Last login time
admin
N/A
N/A
View groups (1)
Admin
22 Oct 26 08:54:35
altayskaya
altayskaya
N/A
View groups (1)
gadukyan
Gadukyan R.A.
N/A
View groups (1)
kaderli
Kaderli Deniz
N/A
View groups (1)

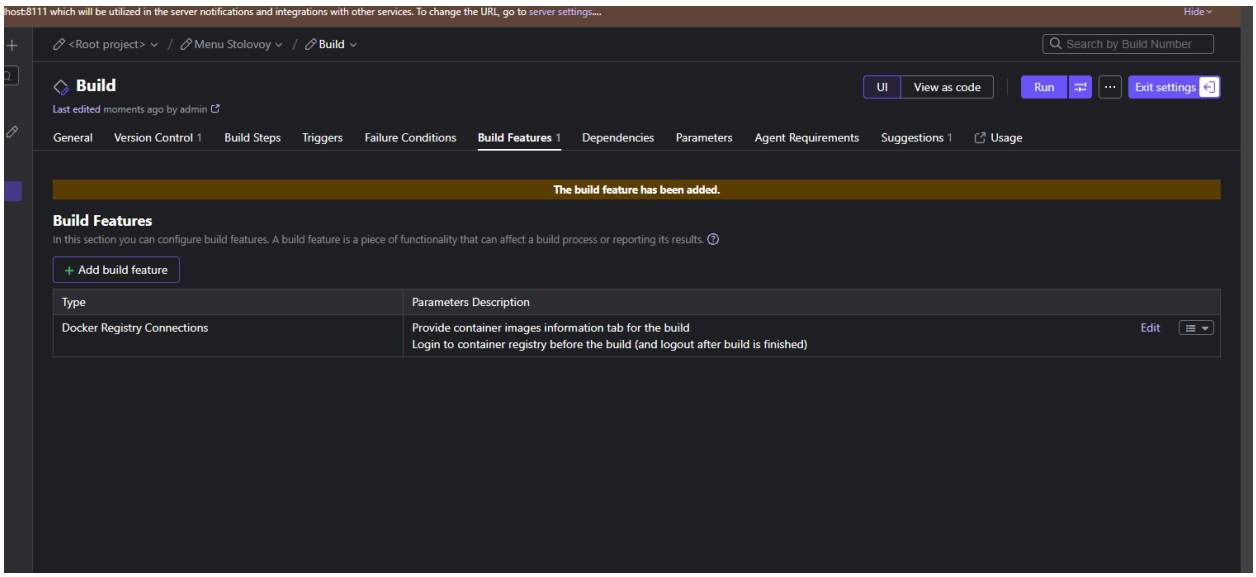

Set up your build
Parent project: Menu Stolovoy
Build configuration
Single build, create and manage chains from the administration or DSL.
Pipeline
Pipeline with multiple jobs, create and configure using visual editor or YAML.
Any Git URL
https://github.com/egrinnn/Automation26.git
SSH
HTTPS
Token
Password
Anonymous
Proceed
Skip

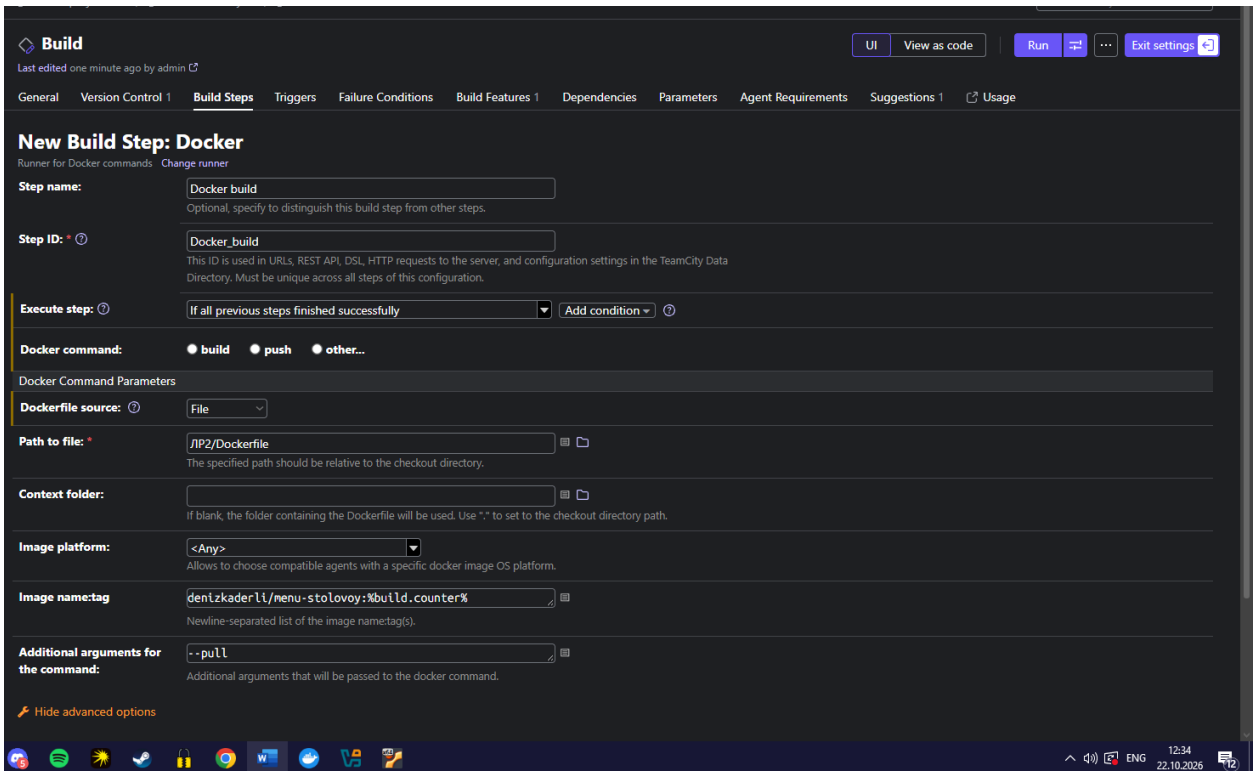

Set up your build
Parent project: Menu Stolovoy
Build configuration
Single build, create and manage chains from the administration or DSL.
Pipeline
Pipeline with multiple jobs, create and configure using visual editor or YAML.
Any Git URL
https://github.com/egrinnn/Automation26.git
Used HTTPS for authentication. Edit authentication settings
Name
Build
Default branch
main
Start builds on new changes in
Branches
All
Default branch
Specific
Show less
Build configuration type
Regular
Description
Create
Skip

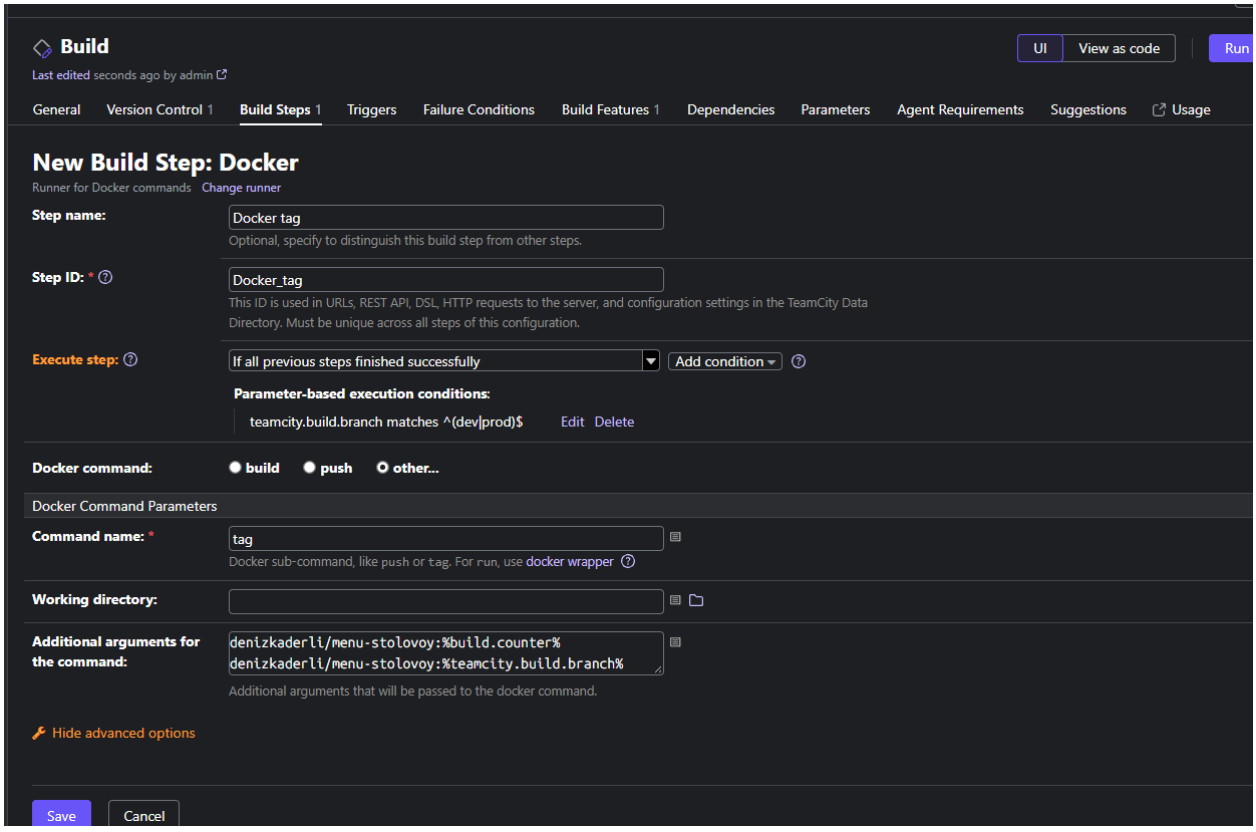

localhost:8111/admin/discoverRunners.html
This server is using the default URL http://localhost:8111 which will be utilized in the server notifications and integrations with other services. To change the URL, go to server settings....
Projects
Settings mode
Search by name
Favorite builds
FAVORITES
Overview
Menu Stolovoy
Build
PROJECTS
Menu Stolovoy
<Root project>
Menu Stolovoy
Build
Build
Created seconds ago by admin
General
Version Control 1
Build Steps
Triggers
Failure Conditions
Build Features
Dependencies
Parameters
Auto-detected Build Steps
Build steps and their settings are detected automatically by scanning VCS repository. You can configure build steps manually if auto-detect did not find relevant build steps.
Build Step
Parameters Description
Docker
Docker build; Dockerfile location: ЛР2/Dockerfile
Python
Command: Pytest
Python
Command: File
File: config.py
Python
Command: File
File: quickstart.py
Python
Command: File
File: run.py
Python
Command: File
File: seed_demo_data.py
Python
Command: File
File: wait_for_db.py
Docker Compose
Start/stop docker-compose from 'ЛР2/docker-compose.yaml'
Command Line
Custom script: ./ЛР2/docker-entrypoint.sh
Use selected
Refresh

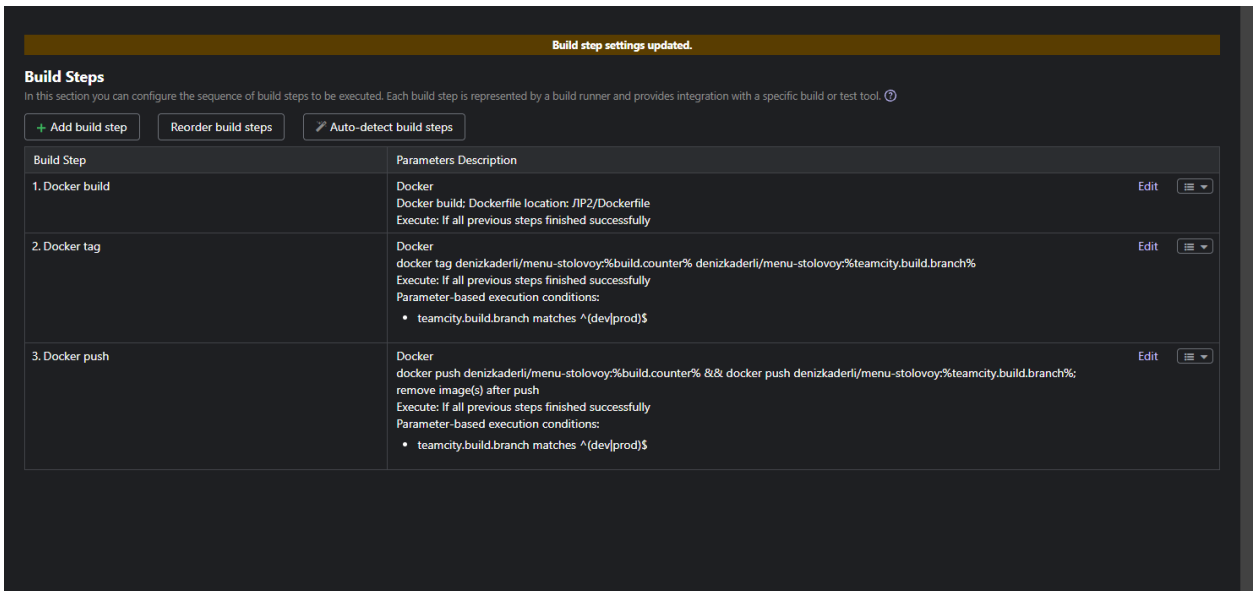

localhost:8111/admin/editProject.html
Menu Stolovoy
Connections
Add Connection
Connection type:
Docker Registry
Test Connection
Connection successful!
Connection to https://registry.hub.docker.com/v2/ was successful!
Save
Cancel
Test Connection

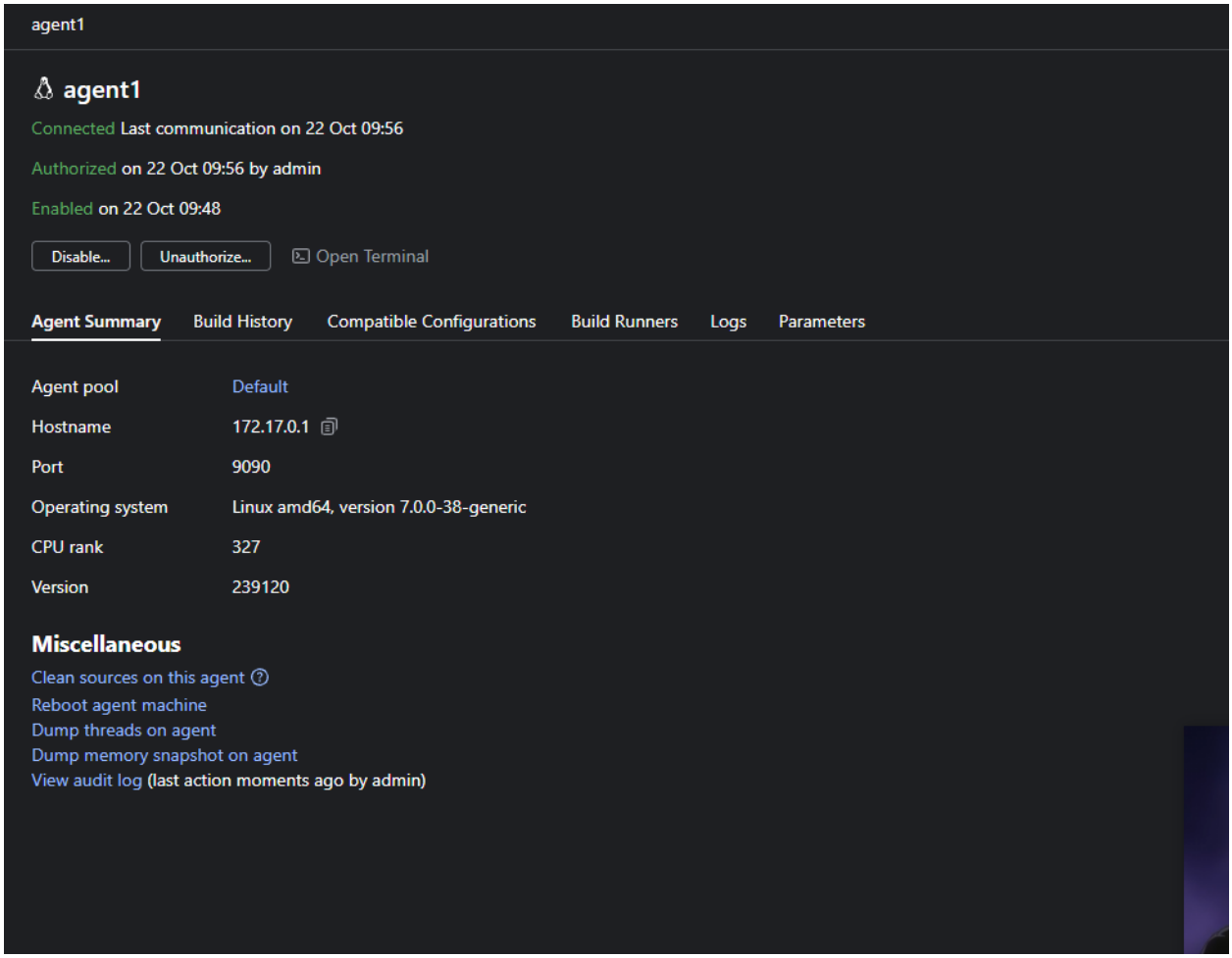

host:8111 which will be utilized in the server notifications and integrations with other services. To change the URL, go to server settings....
Hide
<Root project>
Menu Stolovoy
Search by Build Number
Menu Stolovoy
Created 17 minutes ago by admin
Exit settings
General
VCS Roots 1
Parameters
Recipes
Versioned Settings
Integrations
Connections
Maven Settings
Issue Trackers
Cloud Profiles
Artifacts Storage
Security
VCS Auth Tokens
SSH Keys
Untrusted Builds
Project Isolation
Reporting
Report Tabs
Other
Clean-up Rules
NuGet Feed
Shared Resources
Suggestions 2
Connections
Supported types of connections: Amazon ECR, Amazon Web Services (AWS), Azure DevOps OAuth 2.0, Bitbucket Cloud, Bitbucket Server / Data Center, Docker Registry, GitHub App, GitHub Enterprise OAuth App, GitHub.com OAuth App, GitLab CE/EE, GitLab.com, Google, HashiCorp Vault, Kubernetes Connection, NPM Registry, Perforce Administrator Access, Slack, Azure DevOps PAT, JetBrains Space.
Connection successfully created
+ Add Connection
Connection
Parameters Description
Docker Registry
Connection ID: PROJECT_EXT_2
https://denizkaderli@docker.io
Edit
Delete

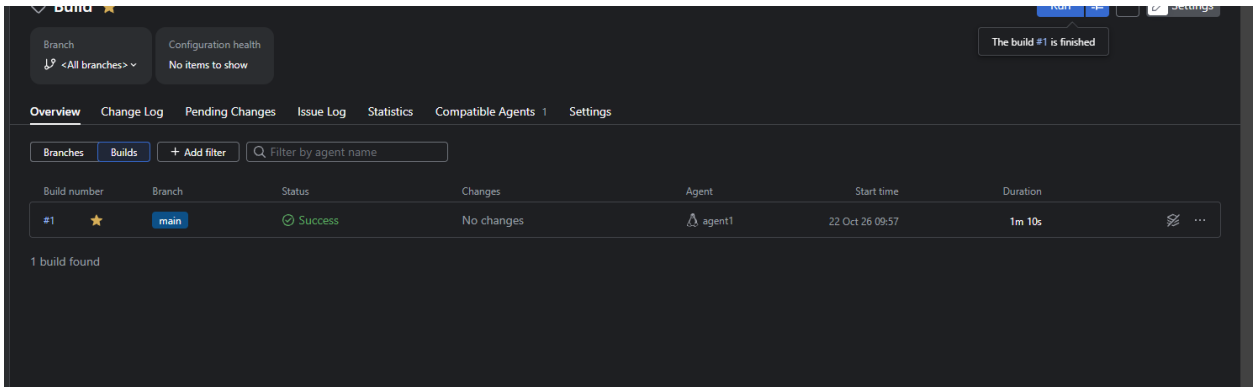

host:8111 which will be utilized in the server notifications and integrations with other services. To change the URL, go to server settings....
Hide
<Root project>
Menu Stolovoy
Build
Search by Build Number
Build
Last edited moments ago by admin
UI
View as code
Run
Exit settings
General
Version Control 1
Build Steps
Triggers
Failure Conditions
Build Features 1
Dependencies
Parameters
Agent Requirements
Suggestions 1
Usage
The build feature has been added.
Build Features
In this section you can configure build features. A build feature is a piece of functionality that can affect a build process or reporting its results.
+ Add build feature
Type
Parameters Description
Docker Registry Connections
Provide container images information tab for the build
Login to container registry before the build (and logout after build is finished)
Edit


Build
Last edited one minute ago by admin
UI
View as code
Run
Exit settings
General
Version Control 1
Build Steps
Triggers
Failure Conditions
Build Features 1
Dependencies
Parameters
Agent Requirements
Suggestions 1
Usage
New Build Step: Docker
Runner for Docker commands
Change runner
Step name:
Docker build
Optional, specify to distinguish this build step from other steps.
Step ID: *
Docker_build
This ID is used in URLs, REST API, DSL, HTTP requests to the server, and configuration settings in the TeamCity Data Directory. Must be unique across all steps of this configuration.
Execute step:
If all previous steps finished successfully
Add condition
Docker command:
build
push
other...
Docker Command Parameters
Dockerfile source:
File
Path to file: *
JIP2/Dockerfile
The specified path should be relative to the checkout directory.
Context folder:
If blank, the folder containing the Dockerfile will be used. Use "." to set to the checkout directory path.
Image platform:
<Any>
Allows to choose compatible agents with a specific docker image OS platform.
Image name:tag
denizkaderli/menu-stolovoy:%build.counter%
Newline-separated list of the image name:tag(s).
Additional arguments for the command:
--pull
Additional arguments that will be passed to the docker command.
Hide advanced options
ENG
12:34
22.10.2026


Build
Last edited seconds ago by admin
UI
View as code
Run
General
Version Control 1
Build Steps 1
Triggers
Failure Conditions
Build Features 1
Dependencies
Parameters
Agent Requirements
Suggestions
Usage
New Build Step: Docker
Runner for Docker commands
Change runner
Step name:
Docker tag
Optional, specify to distinguish this build step from other steps.
Step ID: *
Docker_tag
This ID is used in URLs, REST API, DSL, HTTP requests to the server, and configuration settings in the TeamCity Data Directory. Must be unique across all steps of this configuration.
Execute step:
If all previous steps finished successfully
Add condition
Parameter-based execution conditions:
teamcity.build.branch matches ^(dev|prod)$
Edit
Delete
Docker command:
build
push
other...
Docker Command Parameters
Command name: *
tag
Docker sub-command, like push or tag. For run, use docker wrapper
Working directory:
Additional arguments for the command:
denizkaderli/menu-stolovoy:%build.counter%
denizkaderli/menu-stolovoy:%teamcity.build.branch%
Additional arguments that will be passed to the docker command.
Hide advanced options
Save
Cancel


Build step settings updated.
Build Steps
In this section you can configure the sequence of build steps to be executed. Each build step is represented by a build runner and provides integration with a specific build or test tool.
+ Add build step
Reorder build steps
Auto-detect build steps
Build Step
Parameters Description
1. Docker build
Docker
Docker build: Dockerfile location: ЛР2/Dockerfile
Execute: If all previous steps finished successfully
Edit
2. Docker tag
Docker
docker tag denizkaderli/menu-stolovoy:%build.counter% denizkaderli/menu-stolovoy:%teamcity.build.branch%
Execute: If all previous steps finished successfully
Parameter-based execution conditions:
teamcity.build.branch matches ^(dev|prod)$
Edit
3. Docker push
Docker
docker push denizkaderli/menu-stolovoy:%build.counter% && docker push denizkaderli/menu-stolovoy:%teamcity.build.branch%;
remove image(s) after push
Execute: If all previous steps finished successfully
Parameter-based execution conditions:
teamcity.build.branch matches ^(dev|prod)$
Edit


agent1
agent1
Connected Last communication on 22 Oct 09:56
Authorized on 22 Oct 09:56 by admin
Enabled on 22 Oct 09:48
Disable...
Unauthorize...
Open Terminal
Agent Summary
Build History
Compatible Configurations
Build Runners
Logs
Parameters
Agent pool
Default
Hostname
172.17.0.1
Port
9090
Operating system
Linux amd64, version 7.0.0-38-generic
CPU rank
327
Version
239120
Miscellaneous
Clean sources on this agent
Reboot agent machine
Dump threads on agent
Dump memory snapshot on agent
View audit log (last action moments ago by admin)


Build
Branch
<All branches>
Configuration health
No items to show
The build #1 is finished
Overview
Change Log
Pending Changes
Issue Log
Statistics
Compatible Agents 1
Settings
Branches
Builds
+ Add filter
Filter by agent name
Build number
Branch
Status
Changes
Agent
Start time
Duration
#1
main
Success
No changes
agent1
22 Oct 26 09:57
1m 10s
1 build found